

## Chapter 37. Effect of Mechanical Ventilation on Gas Exchange

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### Effect of Mechanical Ventilation on Gas Exchange: Introduction

The major function of the lung is to exchange physiologic (respiratory) gases, namely **oxygen** ( $O_2$ ) and carbon dioxide ( $CO_2$ ). Once the lungs fail as a gas exchanger, arterial hypoxemia, hypercapnia, or both appear and respiratory failure ensues. Arterial  $P_{O_2}$  ( $Pa_{O_2}$ ) and  $P_{CO_2}$  ( $Pa_{CO_2}$ ) are the measurable end-point variables used routinely by clinicians to manage patients with acute respiratory failure. When the latter is severe, mechanical ventilation is then considered the final strategy for treating patients.

### Relevance

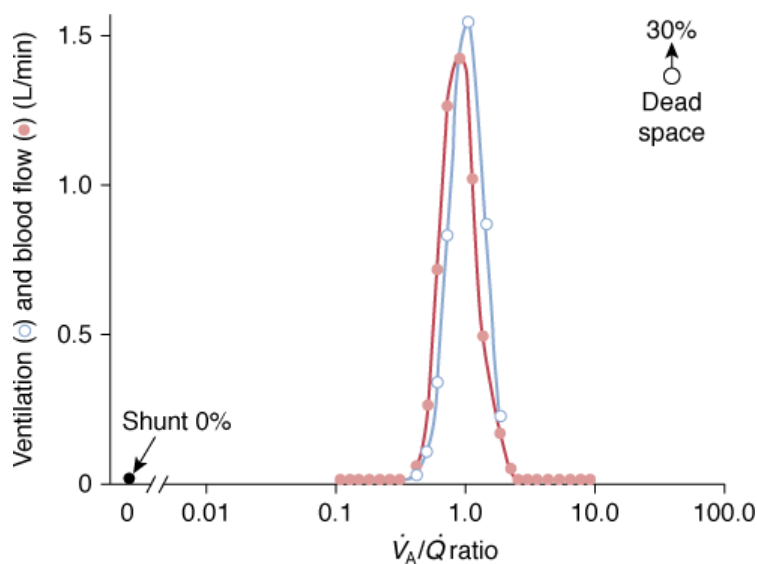
Classically, the mechanisms of hypoxemia are alveolar hypoventilation, limitation of alveolar to end-capillary  $O_2$  diffusion, intrapulmonary shunt, and ventilation–perfusion ( $\dot{V}_A/\dot{Q}$ ) imbalance; the major causes of hypercapnia are alveolar hypoventilation and  $\dot{V}_A/\dot{Q}$  mismatching.<sup>1</sup> Ideally, it would be of great interest to solely manage respiratory blood-gas measurements, such as alveolar-arterial  $P_{O_2}$  difference  $P(A-a)O_2$ , venous admixture ratio ( $\dot{Q}_s/\dot{Q}_t$ ), and physiologic dead space (dead-space-to-tidal-volume ratio [ $V_D/V_T$ ]), as a general marker of the overall function of the lung. Thus, impaired or improved results of these variables, whose principal merits are their simplicity and relative ease of measurement, could reflect impaired or improved pulmonary gas exchange. Unfortunately, all these variables reflect not only the state of the lung, but also the conditions under which the lung is operating. These conditions, which uniquely determine the  $P_{O_2}$  and  $P_{CO_2}$  in any single gas-exchange unit of the lung, are the  $\dot{V}_A/\dot{Q}$  ratio, the composition of inspired gas, and the composition of mixed venous blood heavily modulated by the behavior of the cardiac output.<sup>2,3</sup> It is important to appreciate the key role played by these three factors governing the respiratory gases in any single gas-exchange unit.<sup>3</sup>

### Physiology: Multiple Inert Gas Elimination Technique

This chapter reviews the effect of ventilator support on pulmonary gas exchange using the multiple inert gas elimination technique (MIGET), an approach that represents a major conceptual breakthrough in our understanding of pulmonary medicine pathophysiology in disease states.<sup>4–7</sup> MIGET has three major advantages. First, it estimates the pattern of pulmonary blood flow and alveolar ventilation and calculates the mismatch of  $\dot{V}_A/\dot{Q}$  relationships. Second, it partitions the  $P(A-a)O_2$  into determinants of intrapulmonary shunt,  $\dot{V}_A/\dot{Q}$  inequality, and diffusion limitation to  $O_2$ . Third, it apportions and unravels arterial oxygenation into intrapulmonary, namely the latter three factors, and extrapulmonary components, that is, inspired  $P_{O_2}$ , overall ventilation, cardiac output, and  $O_2$  consumption. Of paramount importance is the ability to perform measurements at any fractional inspired **oxygen** concentration ( $FI_{O_2}$ ) without perturbing the vascular and bronchomotor tones.<sup>8</sup>

Figure 37-1 illustrates the  $\dot{V}_A/\dot{Q}$  distribution obtained with MIGET in a healthy, young individual at rest breathing ambient air. Each data point represents a particular amount of blood flow or alveolar ventilation. Both overall pulmonary perfusion and total ventilation correspond to the sum of the respective data points (the lines have been drawn for clarity only). These quantities (distributions) are plotted against a broad range (50) of  $\dot{V}_A/\dot{Q}$  ratios, from zero (intrapulmonary shunt) to infinity (dead space), on a log scale. The unimodal profile of each distribution has three main characteristics: symmetry, location around a  $\dot{V}_A/\dot{Q}$  ratio of 1, and narrowness (very little dispersion). Note that there is no inert-gas shunt (contrasted with the concept of venous admixture ratio), because the tracer nature of inert gases utilized for MIGET is insensitive to the presence of postpulmonary shunt (i.e., the bronchial and thebesian circulations). Inert-gas physiologic dead space is also slightly lower than Bohr dead space because it does not include the alveolar units, which have an alveolar  $P_{CO_2}$  that is lower than  $Pa_{CO_2}$ .

Figure 37-1



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Distributions of alveolar ventilation and pulmonary blood flow plotted versus  $\dot{V}_A/\dot{Q}$  ratio. For further explanation, see text.

## Nature of Gas Exchange in Disease States

From a clinical standpoint, the three principal mechanisms of altered arterial respiratory gases during spontaneous breathing in any pulmonary disease state are  $\dot{V}_A/\dot{Q}$  mismatching, increased intrapulmonary shunt, and alveolar hypoventilation. The role of diffusion limitation to  $O_2$  is modest and plays a role in patients with pulmonary fibrosis<sup>9</sup> and in healthy individuals under very extreme conditions at high altitude.<sup>10</sup> During mechanical ventilation, however, alveolar hypoventilation is controlled in such a way that  $P_{aCO_2}$  does not represent a problem.  $\dot{V}_A/\dot{Q}$  inequalities play a pivotal role in disorders characterized by chronic lung disease (“dry lung”), namely during exacerbations of chronic obstructive pulmonary disease (COPD) and bronchial asthma, which have in common expiratory airflow limitation and large pulmonary volumes. Increased intrapulmonary shunt is a key determinant of hypoxemia in conditions characterized by acute lung injury (“wet lung”), such as acute lung injury (ALI), and its most severe form, acute respiratory distress syndrome (ARDS), and life-threatening pneumonia, all of which have small lung volumes.<sup>11</sup>

The main mechanisms of pulmonary gas exchange are extensively reviewed (see the section [Acute Respiratory Failure “Wet Lung”](#)) during mechanical ventilation in ALI or ARDS, pneumonia, COPD, and asthma—the most common conditions in the critical care setting. In each condition, the gas-exchange abnormalities at maintenance  $FiO_2$  and while breathing 100%  $O_2$  are addressed. Likewise, the effects of external positive end-expiratory pressure (PEEP) and intrinsic PEEP (PEEPi) on gas exchange, and the effects of several ventilator settings are considered.

### Acute Respiratory Failure (“Wet Lung”)

This section reviews the two most frequent disorders seen in the intensive care unit, ARDS and severe life-threatening pneumonia, together with a short review of gas exchange in patients with head trauma and following cardiac surgery. It can be difficult to differentiate ALI or ARDS from the other disorders—after all, ALI or ARDS is a constellation of many entities, among which pneumonia and the other conditions of acute respiratory failure are common causes. From a gas-exchange viewpoint, however, ALI or ARDS and pneumonia show different functional findings. Furthermore, the response to high  $FiO_2$  differs substantially.

#### Acute Lung Injury or Acute Respiratory Distress Syndrome

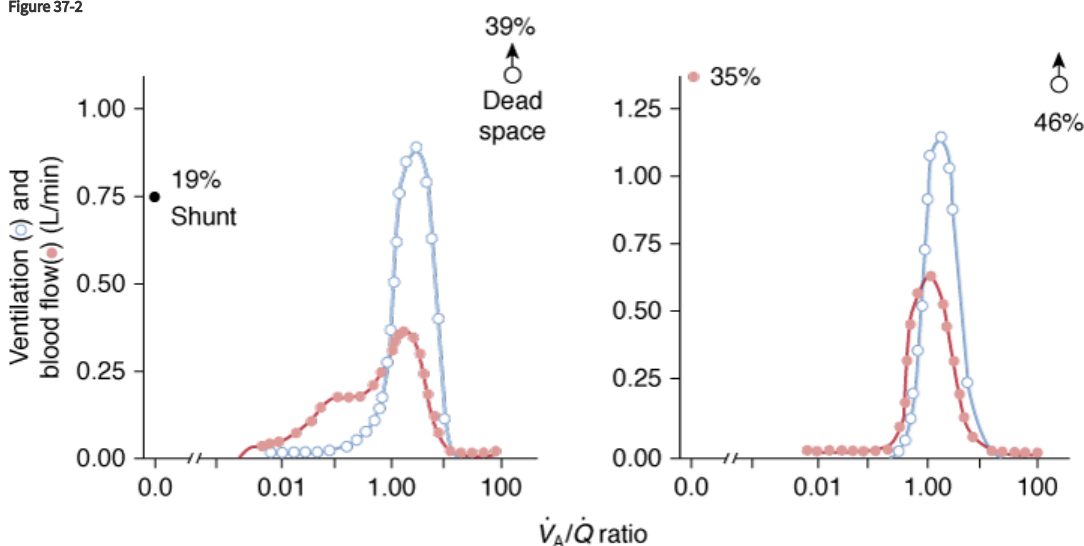
Severe acute respiratory failure in previously healthy subjects may result from a primary infectious lung process or a more widespread, noninfectious process, namely either ALI or ARDS. The latter entity is characterized by severe hypoxemia refractory to high  $FiO_2$ , and differences are related to the severity of gas-exchange abnormalities.<sup>12</sup>

#### Mechanisms of Hypoxemia

The main cause of hypoxemia in patients with ALI or ARDS is characterized by an increased intrapulmonary shunt, averaging 20% or more of cardiac output. In approximately half the patients, however, there are considerable additional areas with low  $\dot{V}_A/\dot{Q}$  ratios: A moderate percentage of total pulmonary blood flow is distributed to areas of the lung with reduced ventilation.<sup>13</sup>

Figure 37-2 illustrates the profiles of two representative  $\dot{V}_A/\dot{Q}$  distributions of patients with ARDS during mechanical ventilation with PEEP (at  $F_{I_{O_2}}$ , 0.5).<sup>14</sup> Although both patterns show marked amounts of shunt and dead space, the main body of the dispersion of blood flow is different. In one patient, the dispersion is narrowly unimodal (*right panel*). In the other, the pattern of perfusion is broadly unimodal, that is, blood flow is distributed to areas with low  $\dot{V}_A/\dot{Q}$  ratios (usually less than 10% of cardiac output) (*left panel*). The former profile reflects an “all-or-none” phenomenon: pulmonary perfusion is essentially diverted to two lung areas, those with ventilation that is normal and proportional to blood flow, and those that are completely unventilated.<sup>15</sup> The presence of areas with low  $\dot{V}_A/\dot{Q}$  ratios suggests the coexistence of areas with partially filled alveolar spaces or alveolar units in which ventilation is reduced compared to blood flow (because of increased airways resistance) or both. A few patients showed an increase of alveolar ventilation distributed to high  $\dot{V}_A/\dot{Q}$  ratios, including a bimodal pattern of the dispersion of ventilation; in contrast, dead space was increased in most patients. These two findings may reflect areas with reduced pulmonary blood flow secondary to the effects of external PEEP, although an additional influence of pulmonary vascular derangement cannot be ruled out.<sup>16</sup>

Figure 37-2

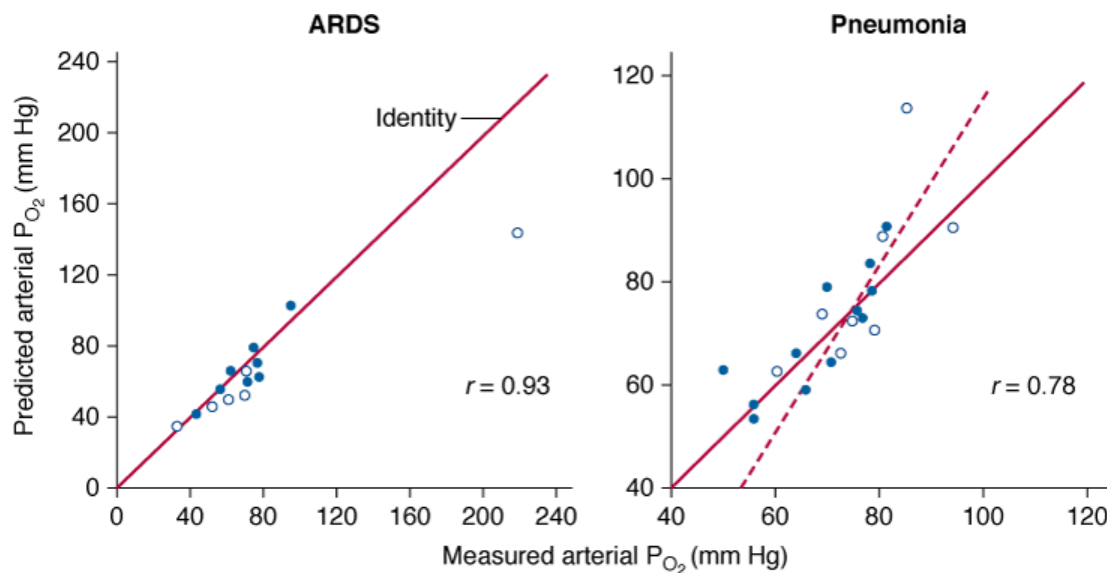


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Two patterns of  $\dot{V}_A/\dot{Q}$  distributions in patients with ARDS (with PEEP). Note that intrapulmonary shunt and dead space were considerably increased in both individuals. Notice also the presence of some amount of blood flow distributed to units with low  $\dot{V}_A/\dot{Q}$  (*left panel*). (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Reyes A, Roca J, Rodriguez-Roisin R, et al. Effect of almitrine on ventilation-perfusion distribution in adult respiratory distress syndrome. *Am Rev Respir Dis*. 1988;137:1062–1067. Official Journal of the American Thoracic Society. Modified with permission of the American Thoracic Society.)

There was no limitation to the diffusion of  $O_2$ , as reflected by the close agreement between predicted  $Pa_{O_2}$  (according to MIGET, this reflects the underlying amount of intrapulmonary shunt and  $\dot{V}_A/\dot{Q}$  mismatch only) and measured  $Pa_{O_2}$  (Fig. 37-3).<sup>15,17</sup> When measured  $Pa_{O_2}$  is not significantly different from the predicted  $Pa_{O_2}$ , this suggests that other potential causes of hypoxemia, namely diffusion limitation for  $O_2$ , increased postpulmonary shunt, and/or augmented intrapulmonary  $O_2$  consumption,<sup>18</sup> are not occurring.

Figure 37-3



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Plots of predicted (estimated by MIGET)  $P_{aO_2}$  versus measured  $P_{aO_2}$  in patients with ARDS (*left panel*; open symbols = without PEEP; closed symbols = with PEEP) and with severe pneumonia (*right panel*; open symbols = with mechanical ventilation; closed symbols = without mechanical support). Note the good agreement between both variables in each clinical condition (*dashed line* = regression line). (Used, with permission, from Dantzker et al.<sup>15</sup> and Gea et al.<sup>17</sup>)

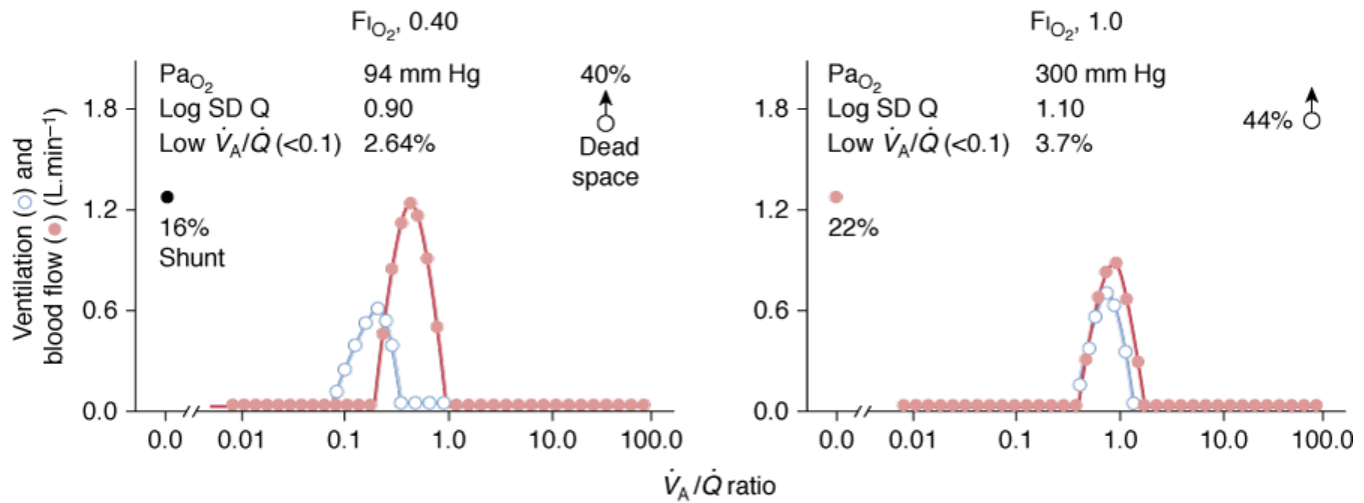
### Breathing 100% Oxygen

The concept of an increase in intrapulmonary shunt while breathing 100% oxygen in patients with acute respiratory failure is old.<sup>19,20</sup> Earlier data using MIGET, however, did not support this view.<sup>16</sup> It was considered that areas with critical inspiratory  $\dot{V}_A/\dot{Q}$  ratios may remain partly open and facilitate  $O_2$  transfer; potential mechanisms include the efficiency of collateral ventilation, interdependence of surrounding lung parenchyma, or the interaction of mechanical forces exerted during artificial ventilation.<sup>8</sup> No studies, however, properly addressed this question.

We have shown that  $P_{aO_2}$  increases modestly ( $\leq 300$  mm Hg) and intrapulmonary shunt increases considerably (approximately 35%) during hyperoxic breathing over 1 hour, suggesting the development of reabsorption atelectasis (Fig. 37-4).<sup>8,21</sup> The increase in intrapulmonary shunt was noticed within less than half an hour, confirming the theoretical analysis on the minimum time required for collapse of alveolar units with various fixed inspired  $\dot{V}_A/\dot{Q}$  ratios according to the different levels of  $F_{IO_2}$ .<sup>8</sup> The limited increase in  $P_{aO_2}$  while breathing 100%  $O_2$  indicates that increased intrapulmonary shunt is the key determinant of hypoxemia in ALI and ARDS. Furthermore, the increments in shunt persisted for 1 hour after resuming maintenance  $F_{IO_2}$ , indicating the persistence of atelectasis. Of note, this worsening of shunt was also accompanied by a small but significant increase in  $P_{aCO_2}$ , possibly explained by the Haldane effect (Fig. 37-5). The Haldane effect is the increase in  $P_{aCO_2}$  at a given arterial  $CO_2$  content that occurs in response to an increase in arterial  $O_2$  saturation. The dispersions of blood flow and alveolar ventilation remained unchanged.

Figure 37-4

## ARDS

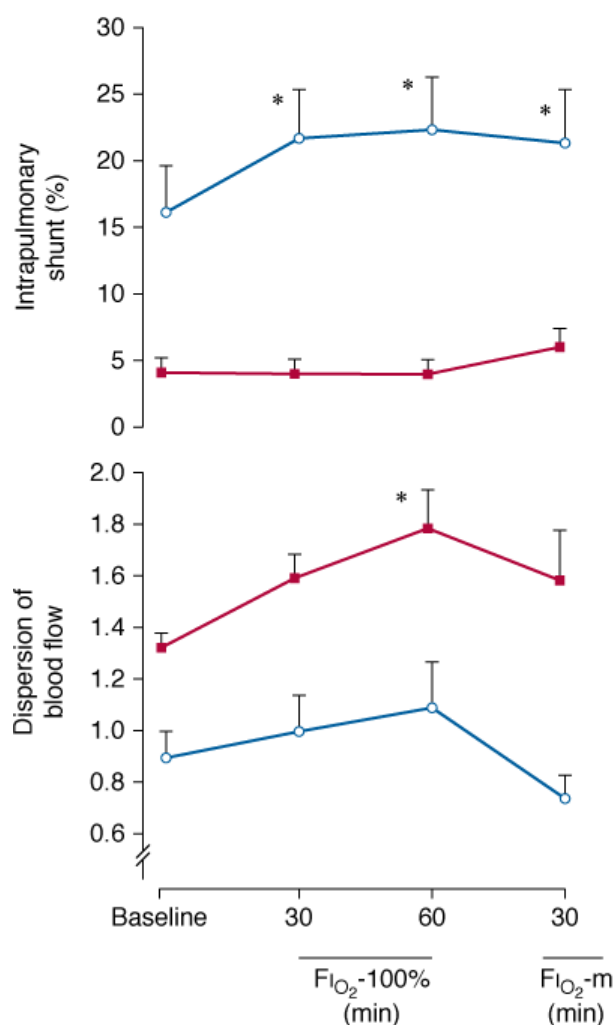
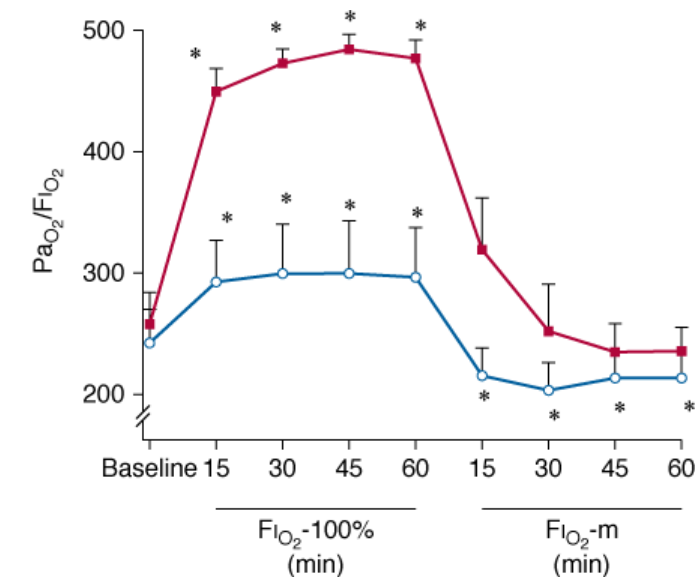


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Effect of breathing 100% oxygen (right panel) in a representative patient with ARDS (with PEEP). Compared with low FiO<sub>2</sub> (left panel), PaO<sub>2</sub> increased modestly (approximately 300 mm Hg), but intrapulmonary shunting increased moderately; in contrast, both the dispersion of blood flow (Log SD Q) and the amount of blood flow distributed to areas with low  $\dot{V}_A/\dot{Q}$  ratios remained unchanged. This suggests the development of reabsorption atelectasis without release of hypoxic vasoconstriction. Dead space tended to increase.

Figure 37-5



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Sequence of PaO<sub>2</sub>/F<sub>i</sub>O<sub>2</sub> (in millimeters of mercury), shunt (in percentage of cardiac output), and Log SD Q (dimensionless). In patients with ALI (*open circles*) arterial oxygenation and Log SD Q remained almost unchanged, whereas shunt increased significantly, suggesting reabsorption atelectasis. In contrast, in patients with COPD (*solid squares*), Log SD Q and arterial oxygenation substantially increased whereas shunt remained low and unvaried, indicating release of hypoxic vasoconstriction. For further explanation, see text. (Used, with permission, from Santos et al.<sup>21</sup>)

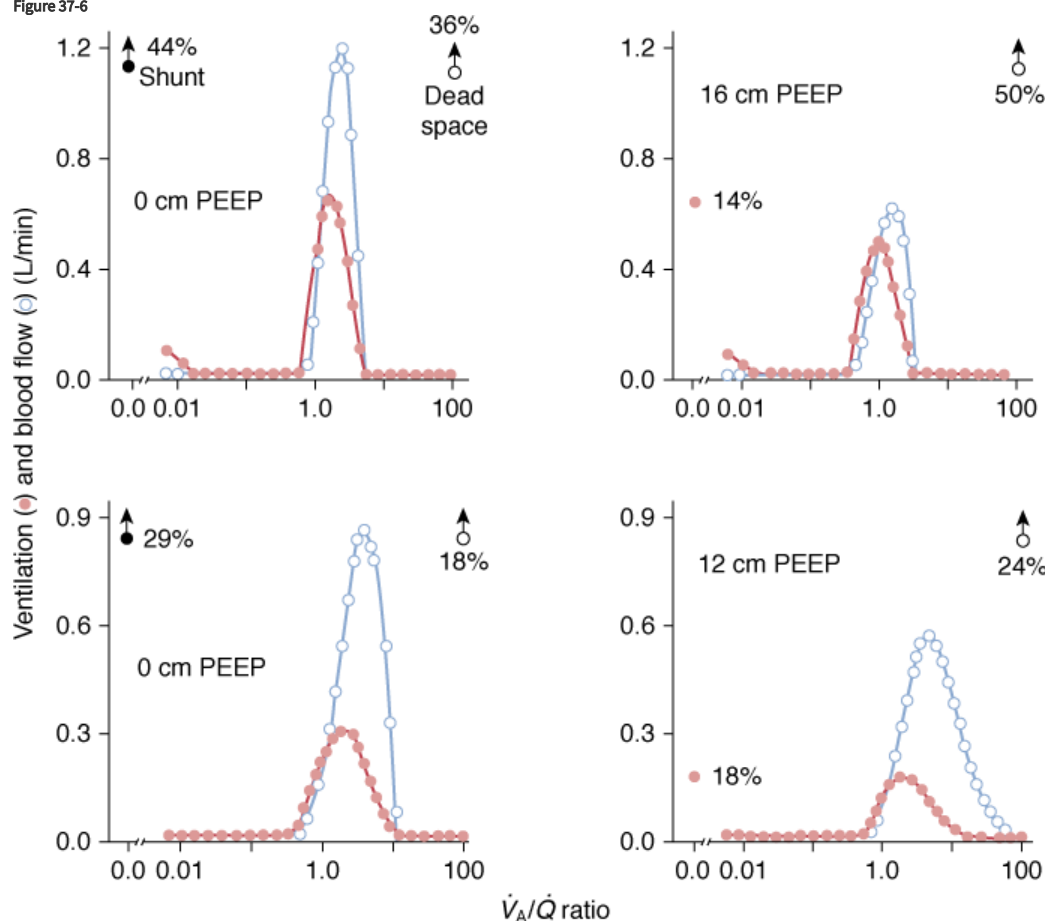
The increase in shunt not accompanied with release of hypoxic pulmonary vasoconstriction in patients with ALI or ARDS is compatible with the concept that, at any level of  $\text{FI}_{\text{O}_2}$ , units with low  $\dot{V}_A/\dot{Q}$  cannot redistribute blood flow from areas of intrapulmonary shunt or very low  $\dot{V}_A/\dot{Q}$  ratios, because their vascular resistance remains unchanged.<sup>8</sup>

### Effects of Positive End-Expiratory Pressure

The gas-exchange response to external PEEP illustrates one of the best examples of the complex interplay between intrapulmonary and extrapulmonary determinants of respiratory gases.<sup>5,16</sup> Several studies have assessed this action in patients with ALI or ARDS.

In a seminal study,<sup>15</sup> which measured  $\dot{V}_A/\dot{Q}$  distributions at increasing levels of PEEP, two distinct patterns were shown (Fig. 37-6). In some patients, the  $\dot{V}_A/\dot{Q}$  pattern remained unchanged despite progressive increases of PEEP. In others, there was a broadening of the dispersion of ventilation because areas with high  $\dot{V}_A/\dot{Q}$  ratios increased and ventilation was redistributed to infinity  $\dot{V}_A/\dot{Q}$  ratios. In each patient, however, PEEP led ultimately to a marked decrement in cardiac output distributed to unventilated (shunt) areas or poorly ventilated  $\dot{V}_A/\dot{Q}$  units and to a considerable increase in dead space. The final result was a substantial optimization of  $\text{PaO}_2$ .

Figure 37-6



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The application of 16 cm  $\text{H}_2\text{O}$  of PEEP in a patient with ARDS induced a reduction of shunt and an increase of dead space, whereas  $\dot{V}_A/\dot{Q}$  distributions remain essentially unaltered (*top, from left to right*). In another patient, 12 cm  $\text{H}_2\text{O}$  of PEEP caused similar effects on shunt and dead space, but  $\dot{V}_A/\dot{Q}$  distributions are broadened (*bottom, from left to right*). For further explanation, see text. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Dantzker DR, Brook CJ, Dehart P, et al. Ventilation-perfusion distributions in the adult respiratory distress syndrome. *Am Rev Respir Dis*. 1979;120:1039–1052. Official Journal of the American Thoracic Society. Modified with permission of the American Thoracic Society.)

Subsequently, the  $\dot{V}_A/\dot{Q}$  response to external PEEP was investigated while cardiac output was kept constant at control values with dopamine.<sup>22</sup> During PEEP,  $\text{PaO}_2$ , mixed venous  $\text{P}_{\text{O}_2}$ , and oxygen delivery increased significantly, whereas the venous admixture ratio decreased markedly.

Similarly, shunt decreased substantially, and pulmonary blood flow was redistributed from nonventilated units with zero  $\dot{V}_A/\dot{Q}$  ratios to areas with normal  $\dot{V}_A/\dot{Q}$  ratios. Dead space increased slightly, but the dispersion of the alveolar ventilation did not vary. Because PEEP causes redistribution of extravascular lung water from alveoli to peribronchial and perivascular spaces,<sup>23</sup> these data suggest that the improvement in pulmonary gas exchange with external levels of PEEP results from the reopening of collapsed airways and alveoli. The key message from this study is that the avoidance of the reduction of cardiac output during PEEP application enhances a more optimal gas exchange.

In another study,<sup>24</sup> the gas-exchange response to increments in PEEP was studied. In most patients,  $\text{PaO}_2$  improved substantially because of a decrease in intrapulmonary shunt or in the zone of low  $\dot{V}_A/\dot{Q}$  ratio, or a derecruitment of blood flow from areas with shunt to those with low or normal  $\dot{V}_A/\dot{Q}$  ratio; blood flow reductions to low  $\dot{V}_A/\dot{Q}$  areas were less predictable. When  $\text{PaO}_2$  did not vary in response to PEEP, there were no variations in the  $\dot{V}_A/\dot{Q}$  relationships. An increase either in poorly perfused  $\dot{V}_A/\dot{Q}$  units or in dead space was shown in only a few individuals. The beneficial effect of PEEP on  $\text{PaO}_2$  could not be predicted by the etiology of ARDS or the severity of abnormal gas exchange. As expected, cardiac output and mixed venous  $\text{P}_{\text{O}_2}$  decreased progressively as PEEP was increased. The mean change in cardiac output was essentially similar between PEEP trials irrespective of the improvement in  $\text{PaO}_2$ .

Experimentally, low levels of PEEP (5 to 10 cm H<sub>2</sub>O) decrease physiologic dead space by reducing both shunt and  $\dot{V}_A/\dot{Q}$  mismatch.<sup>25</sup> At higher levels of PEEP, however, physiologic dead space increases because both ventilation to high  $\dot{V}_A/\dot{Q}$  regions and anatomic dead space increase while the efficiency of CO<sub>2</sub> elimination by the lungs is diminished.

All in all, these studies are of interest for two reasons. First, these findings indicate that the beneficial effects of PEEP in ALI and ARDS essentially result from a redistribution of blood flow from severely injured (shunt) areas to poorly or normally ventilated alveolar units. This seems to be related to the reopening of collapsed alveoli and airways rather than to a decrease of cardiac output by itself. Alternatively, the depression of cardiac output may reduce shunt or blood flow distributed to units with poorly ventilated  $\dot{V}_A/\dot{Q}$  ratios.<sup>2</sup> It has been shown both clinically and experimentally, using pharmacologic and mechanical techniques, that changes in cardiac output lead to directionally similar changes in shunt. Likewise, in many instances, the shunt fraction also changes in a similar direction to changes in the mixed venous and negatively with pulmonary vascular resistance.<sup>26</sup>

Second, in two of the studies in which the values of mixed venous  $\text{P}_{\text{O}_2}$  were reported, this variable either increased<sup>22</sup> or remained stable<sup>24</sup> when cardiac output was reduced. Arterial  $\text{P}_{\text{O}_2}$  increased during PEEP because the beneficial effects on gas exchange, essentially secondary to the reduction in shunt (intrapulmonary factor), were not offset by the simultaneous deleterious effect on  $\text{PaO}_2$  secondary to decreased cardiac output (extrapulmonary factor), which had allowed mixed venous  $\text{P}_{\text{O}_2}$  to decrease, other factors being equal. If the reduction in cardiac output had been more severe,  $\text{PaO}_2$  would have remained unaltered or even decreased despite the reduction in shunt.<sup>16</sup> Alternatively, the observation that mixed venous  $\text{P}_{\text{O}_2}$  values did not always vary in the same direction as cardiac output raises the question of the pathologic supply dependence between O<sub>2</sub> delivery and the O<sub>2</sub> consumption.<sup>27</sup> Such patients may respond to the depression of cardiac output during PEEP by reducing O<sub>2</sub> consumption.

The development of high  $\dot{V}_A/\dot{Q}$  peaks and the moderate-to-severe increments of dead space during PEEP are consistent with experimental data indicating that the reduction of cardiac output and the depression of perfusion in the uppermost regions of the lung determine changes in the pattern of ventilation with PEEP.<sup>28</sup>

## Ventilator Modalities

Over the last 10 years, increasing emphasis has been placed on avoiding ventilator-associated lung injury and the use of a protective ventilator strategy, accepting an increase in  $\text{Pa}_{\text{CO}_2}$  if necessary (permissive hypercapnia), in patients with ALI or ARDS. Ventilator-associated lung injury<sup>29</sup> is defined as lung damage with characteristic features of ARDS that can occur in patients receiving mechanical ventilation. The two most important factors responsible for ventilator-associated lung injury are thought to be: (a) the association of alveolar overdistension and high transpulmonary pressure; and (b) repeated alveolar collapse and reopening owing to ventilation at inappropriate tidal ranges of transpulmonary pressure. The prevention of alveolar overdistension by reducing tidal volume has been investigated in combination with relatively low levels of PEEP and high levels of PEEP. We describe them separately.



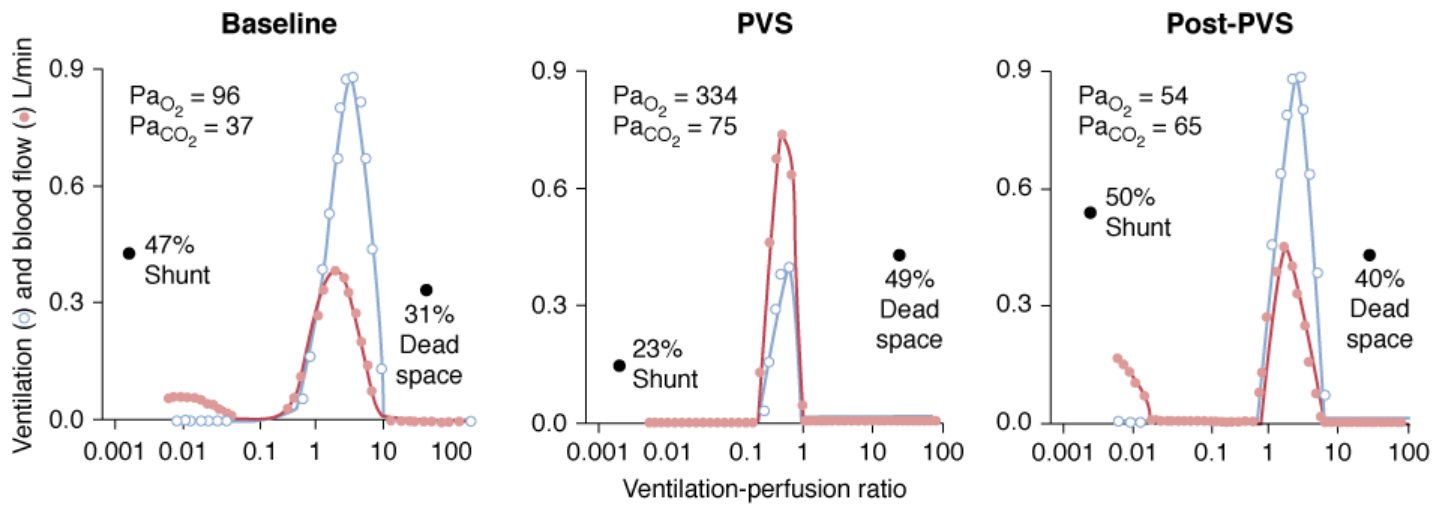
Four recent controlled studies<sup>13,30–32</sup> have compared the outcome with use of low (6 to 7 mL/kg) versus high (10 to 12 mL/kg) tidal volumes with standard levels of PEEP (8 to 11 cm H<sub>2</sub>O). Only one study<sup>13</sup> showed a significant (20%) decrease in mortality, likely because of a superior statistical power and a larger difference between the low-tidal-volume and high-tidal-volume branches. Use of low tidal volume, however, can worsen pulmonary gas exchange, necessitating higher FI<sub>O2</sub> requirements, secondary to reduction in alveolar ventilation and increased shunt caused by both alveolar collapse and increased cardiac output, which may further worsen intra-pulmonary shunt in patients with ARDS.<sup>15</sup>

The induction of permissive hypercapnia by reducing tidal volume and maintaining constant standard levels of PEEP was shown to increase cardiac output, decrease PaO<sub>2</sub>, and increase shunt, but had no effect on the dispersion of blood flow (Log SD Q).<sup>33</sup> When baseline tidal volume was reinstated and cardiac output was maintained with use of dobutamine, both shunt and PaO<sub>2</sub> remained unchanged. Thus, low tidal volume with standard PEEP increased intrapulmonary shunt and the deterioration in gas exchange was caused by the combined effects of increased cardiac output and decreased alveolar ventilation. In another study,<sup>34</sup> the effects of permissive hypercapnia were compared in two groups of patients with ARDS, with and without hyperdynamic sepsis. Permissive hypercapnia was induced by decreasing tidal volume and increasing PEEP to prevent alveolar recruitment. In both groups of patients, permissive hypercapnia increased shunt but did not worsen PaO<sub>2</sub> because of an increase in mixed venous P<sub>O2</sub> secondary to an increased cardiac output. In septic patients, PaO<sub>2</sub> remained unchanged, and in nonseptic patients, it increased. Dead space and the descriptors of  $\dot{V}_A/\dot{Q}$  imbalance remained unchanged in both subsets of patients.

#### **Permissive Hypercapnia with Low Tidal Volume and High Positive End-Expiratory Pressure.**

The combination of low tidal volume, to avoid alveolar overdistension, and high PEEP, to prevent sequential collapse and reopening of alveolar units throughout the ventilatory cycle, has been advocated as an alternative strategy to decrease ventilator-associated lung injury. In patients with ARDS,<sup>35</sup> this approach achieved a significant reduction in mortality at 4 weeks, but hospital mortality was not decreased. The protective ventilation group had a higher rate of weaning from mechanical ventilation and a lower rate of barotrauma. This type of permissive hypercapnia produced an acute hyperdynamic state,<sup>36</sup> with transient systemic vasodilation and increased cardiac output. The acute hemodynamic effects were in part related to acute respiratory acidosis, and progressively attenuated during the first 36 hours of ventilation despite persisting hypercapnia. The gas-exchange response to this ventilator strategy<sup>37</sup> revealed superior efficacy in early ARDS (Fig. 37-7), mainly through recruitment of previously collapsed alveoli and redistribution of pulmonary blood flow from unventilated alveolar units to normal units. Arterial P<sub>O2</sub> improved and intrapulmonary shunt decreased. Recruitment of previously collapsed alveolar units was the key factor underlying the decreased intrapulmonary shunt. The beneficial effects of this approach on gas exchange fully overcame the deleterious effect of permissive hypercapnia on arterial oxygenation (i.e., reduced alveolar P<sub>O2</sub>). Moreover, the increase in cardiac output secondary to systemic vasodilation<sup>36</sup> did not result in a proportional increase in intrapulmonary shunt, as one might expect in patients with ARDS,<sup>15</sup> likely because pulmonary blood flow was appropriately redistributed to normal alveolar units secondary to the concomitant efficiency of alveolar recruitment (Fig. 37-8). The substantial improvement in  $\dot{V}_A/\dot{Q}$  imbalance, indicated by decreased blood flow dispersion (Log SD Q), may have been caused by the reduction of the overall  $\dot{V}_A/\dot{Q}$  ratio secondary to the concomitant effects of both decreased alveolar ventilation and increased cardiac output (induced by this ventilator modality). Under these circumstances, a more efficient distribution of pulmonary blood flow toward areas with normal  $\dot{V}_A/\dot{Q}$  ratios cannot be ruled out. Likewise, the marked increase in dead space observed with the protective strategy can be explained by the combination of a decrease in alveolar ventilation and an increase in functional residual capacity induced by high levels of PEEP. This study showed a strong relationship between improvement of alveolar gas exchange and the amount of PEEP-induced alveolar recruitment. Overall, these findings strongly support the combination of low tidal volumes and high PEEP levels as the most appropriate approach, at least in early ARDS, although further studies are warranted. The growing knowledge that therapeutic hypercapnia per se may prevent pulmonary and systemic damage in patients with ARDS<sup>38,39</sup> adds further complexities.

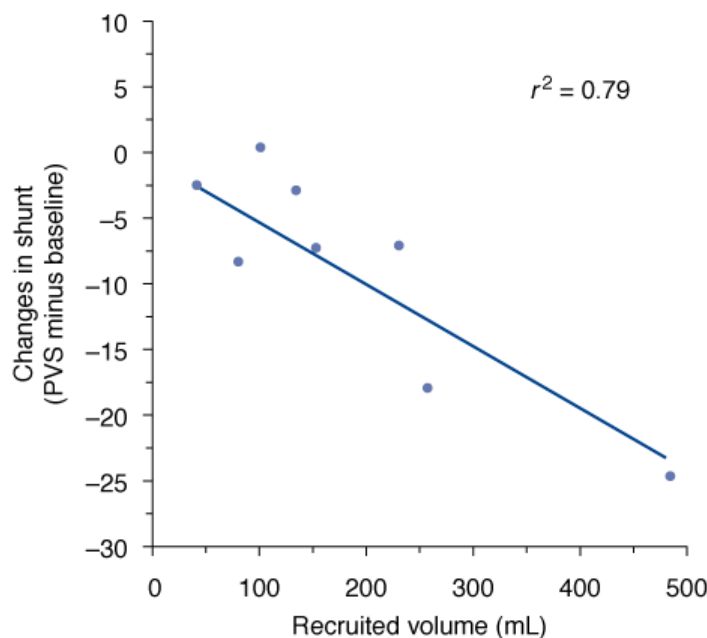
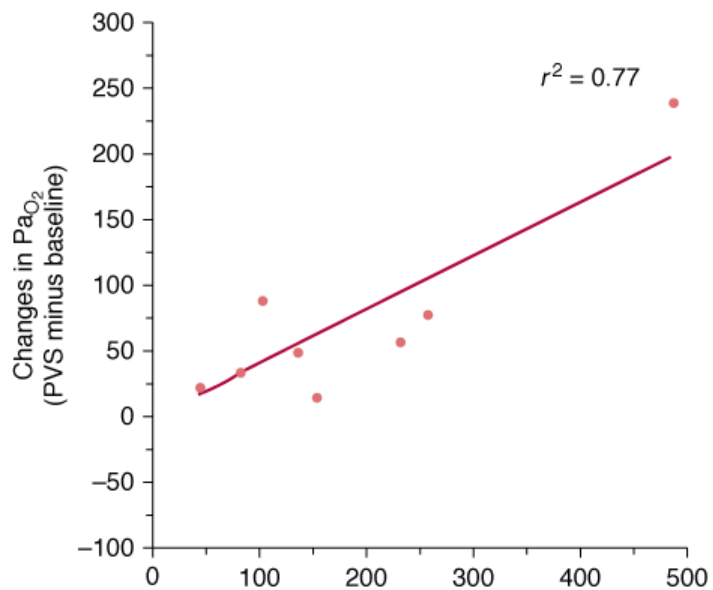
Figure 37-7



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Representative patient with ARDS at baseline (*left*), during protective ventilator support (PVS) (*center*), and after PVS (*right*). During PVS, shunt decreased and arterial oxygenation improved; dead space increased as a result of low tidal volume and high PEEP. (Reproduced, with permission, from Mancini et al.<sup>37</sup>)

Figure 37-8



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Correlations between increased PaO<sub>2</sub> (*top*) and decreased shunt (*bottom*) and recruited lung volume during protective ventilator support (PVS) in patients with ARDS (*dots*, individual patients; *lines*, regression lines). (Reproduced, with permission, from Mancini et al.<sup>37</sup>)

#### Inverse Inspiratory-to-Expiratory Ratio Ventilation.

Before the advantages of protective ventilation with permissive hypercapnia were known, inverse inspiratory-to-expiratory (I:E) ratio ventilation was used in ARDS to improve oxygenation at lower-than-conventional peak airway pressures. Because of discrepant results in several studies, the gas-exchange response was investigated.<sup>40</sup> Compared to conventional volume-controlled ventilation with PEEP, the application of equivalent levels of PEEP during inverse-ratio ventilation did not result in a superior PaO<sub>2</sub>. Pressure-controlled ventilation with inverse I:E ratio did not affect PaO<sub>2</sub> either, although it decreased Pa<sub>a</sub>CO<sub>2</sub> because of a concomitant decrease in dead space with a shift to the right of  $\dot{V}_A/\dot{Q}$  distributions.

#### Postural Changes.

The  $\dot{V}_A/\dot{Q}$  distributions during pressure-controlled ventilation in the prone position<sup>41</sup> revealed an improvement in PaO<sub>2</sub> of more than 10 mm Hg after 30 minutes in two-thirds of patients (responders), whereas the remaining patients showed no change in PaO<sub>2</sub> (nonresponders). In the

responders, the prone position caused a substantial decrease in shunt regions with a concomitant increase of areas with normal  $\dot{V}_A/\dot{Q}$  ratios in the dorsal zones, which were now less gravitationally dependent. Returning the patient to the supine position reversed the improvement in gas exchange. These changes suggest that the redistribution of blood flow away from shunt areas is most likely caused by efficient recruitment of previously atelectatic, but nondiseased, zones induced by altered gravitational forces. Continuous axial rotation might be another method for acutely reducing  $\dot{V}_A/\dot{Q}$  imbalance in patients with ALI or ARDS<sup>42</sup>; continuous axial rotation on a kinetic-treatment table reduced intrapulmonary shunt and improved  $\text{PaO}_2$ . The positive response to continuous rotation was only demonstrated in patients with mild-to-moderate ALI; in patients with progressive ARDS, the acute response was limited.

#### Partial Ventilator Support.

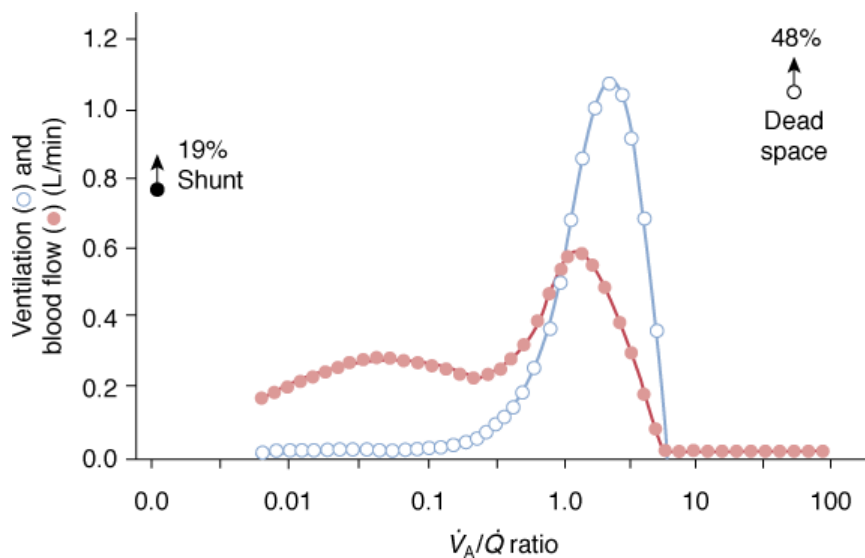
A study comparing the effects of airway pressure release ventilation, with and without spontaneous breathing, and pressure support ventilation, in which both modalities were delivered with equal airway pressure limits or minute ventilation, on pulmonary gas exchange in patients with ARDS,<sup>43</sup> showed that spontaneous breathing with airway pressure release ventilation improved  $\dot{V}_A/\dot{Q}$  mismatching, as revealed by decreases in intrapulmonary shunt, dead space, and the dispersions of blood flow and ventilation. Pressure support did not improve  $\dot{V}_A/\dot{Q}$  imbalance when compared with airway pressure release ventilation without spontaneous breathing. These findings indicated that uncoupling of spontaneous and mechanical ventilation during airway pressure release ventilation improves  $\dot{V}_A/\dot{Q}$  inequalities in ARDS, presumably by recruiting nonventilated lung units. As compared with controlled ventilation, inspiratory assistance during pressure support is not sufficient to counteract the  $\dot{V}_A/\dot{Q}$  imbalance caused by alveolar collapse in patients with ARDS, and does not provide any complementary advantage in cardiopulmonary function or pulmonary gas exchange.

### Pneumonia

#### Mechanisms of Hypoxemia.

In pneumonia, arterial hypoxemia is basically determined by the presence of a considerable shunt (average: 20% of cardiac output) together with moderate to severe amounts (range: 10% to 20% of cardiac output) of blood flow distributed to units with low  $\dot{V}_A/\dot{Q}$ . In general, the pattern of the dispersion of perfusion is bimodal (Fig. 37-9).<sup>17,44</sup> These functional findings are consistent with the main pathologic findings, namely extensive consolidation of areas of the lungs, with alveoli completely or partially filled with edema, leukocytes and other cells, causative bacteria, and fibrin.<sup>45</sup> Dead space is, by and large, slightly increased, and some patients can exhibit areas wherein ventilation is diverted to high  $\dot{V}_A/\dot{Q}$  ratios. The latter findings may reflect the use of PEEP, which is commonly used in these patients. There is good agreement between predicted  $\text{PaO}_2$  and measured  $\text{PaO}_2$ , suggesting therefore that additional intrapulmonary factors determining hypoxemia (diffusion limitation to  $\text{O}_2$ , increased intrapulmonary  $\text{O}_2$  uptake, and postpulmonary shunt), other than shunt and  $\dot{V}_A/\dot{Q}$  mismatch, can be ruled out.<sup>17,44</sup> The coexistence of increased intrapulmonary  $\text{O}_2$  consumption within the parenchyma<sup>18</sup> and increased postpulmonary shunt,<sup>46</sup> previously suggested as an additional mechanism contributing to hypoxemia, was not observed in patients with severe pneumonia.<sup>17</sup>

Figure 37-9



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Ventilation-perfusion distributions in a patient with severe pneumonia requiring mechanical ventilation with PEEP. Note that shunt and dead space are markedly increased; likewise, a considerable amount of blood flow is distributed to a zone with low  $\dot{V}_A/\dot{Q}$  ratios (below 0.1). Compared with patients with ARDS (see Fig. 37-2), the bimodal pattern of pulmonary perfusion is more accentuated. (Reproduced, with permission, from Gea et al.<sup>17</sup>)

#### Breathing 100% Oxygen.

Breathing 100% O<sub>2</sub> induced no changes in inert-gas shunt or in venous admixture ratio, as might be expected, at least theoretically, in patients whose lungs contain abundant critical alveolar units at risk of undergoing collapse, thereby developing reabsorption atelectasis.<sup>8,21</sup> This suggests that, in lungs with more localized lung injury, low  $\dot{V}_A/\dot{Q}$  areas are less liable to collapse during oxygen breathing, probably because of the efficiency of collateral ventilation and the interaction between mechanical forces within the lung. Nevertheless, there was further  $\dot{V}_A/\dot{Q}$  deterioration, as assessed by the marked increase in the dispersion of blood flow, indicating release of hypoxic pulmonary vasoconstriction. This finding indicates a considerable hypoxic vascular response of the lung, which may play a protective role against further worsening of gas exchange. In contrast, the dispersion of ventilation tended to decrease (improve) and dead space remained unaltered (Table 37-1). Conceivably, this mild reduction in the dispersion of alveolar ventilation might reflect blood flow redistribution from high- to low-  $\dot{V}_A/\dot{Q}$  regions, where hypoxic vasoconstriction takes place.

Table 37-1: Characteristics of Pulmonary Gas Exchange in Acute Lung Injury “Wet Lung”

|                                     | ALI or ARDS                       | Pneumonia                  |
|-------------------------------------|-----------------------------------|----------------------------|
| <i>Principal mechanisms</i>         |                                   |                            |
| Shunt                               | Severe $\geq 20\%$                | Severe $\geq 20\%$         |
| $\dot{V}_A/\dot{Q}$ mismatch        | Absent/mild                       | Mild/moderate              |
| O <sub>2</sub> diffusion limitation | Absent                            | Absent                     |
| <i>100% Oxygen effects</i>          |                                   |                            |
| Increase in PaO <sub>2</sub>        | Mild/moderate ( $\leq 300$ mm Hg) | Marked ( $\geq 300$ mm Hg) |
| Increase in shunt                   | Mild/moderate                     | Absent                     |
| Hypoxic vascular response           | Absent                            | Increased                  |

#### Postural Changes.

Ventilation–perfusion distributions have been studied in a few patients with unilateral lung disease (presence of infiltrates) needing mechanical ventilation.<sup>47</sup> Two patients displayed a predominant decrease in intrapulmonary shunt and two other patients showed an improvement in  $\dot{V}_A/\dot{Q}$  distributions when the uninvolved lung side was dependent. Although this difference in response may be related to the lack of clinical uniformity in the small number of patients studied, both responses resulted in increases in PaO<sub>2</sub> when the healthy lung was dependent.

#### Head Trauma

Patients with head trauma with normal chest radiology and without clinical neurogenic pulmonary edema show mild-to-severe  $\dot{V}_A/\dot{Q}$  mismatch, essentially characterized by the presence of areas of low  $\dot{V}_A/\dot{Q}$  ratios and mild shunt (less than 10% of cardiac output) (without PEEP). Although no satisfactory explanation has been given for such gas-exchange abnormalities,<sup>48</sup> subclinical pulmonary edema on days before admission, atelectasis, and/or decreased lung compliance have been suggested as potential mechanisms.

When mechanical ventilation was discontinued, shunt remained almost unchanged, while  $\dot{V}_A/\dot{Q}$  distributions were less homogeneous and less unimodal, with a substantial increase in perfusion to regions of low  $\dot{V}_A/\dot{Q}$  ratio units. Functional residual capacity, minute ventilation, and cardiac output were no different between both conditions, but pulmonary arterial pressures increased significantly when patients were removed from the ventilator. Two patients improved dramatically in a few days with considerable, although incomplete, restoration of  $\dot{V}_A/\dot{Q}$  mismatch to normal. This suggested the return of normal hypoxic pulmonary vasoconstriction.

#### Cardiac Surgery

Gas exchange has been studied in patients on the day after coronary bypass surgery for myocardial revascularization<sup>49–51</sup> or aortic valvular replacement.<sup>52</sup> Overall, shunt was mildly to moderately increased (less than 15% to 20% of cardiac output) and  $\dot{V}_A/\dot{Q}$  distributions showed a variable degree of severity, in general akin to preoperative findings, with variable patterns (broadly unimodal or clearly bimodal) of  $\dot{V}_A/\dot{Q}$  abnormalities. The presence of relatively abundant shunt was also consistent with the presumed underlying multifactorial pathology in these patients. Shunt has been attributed to noncardiogenic pulmonary edema related to cardiopulmonary bypass, reduced functional residual capacity, surfactant alterations, increased closing volume secondary to retained secretions, abnormal hypoxic pulmonary vasoconstriction, and reabsorption atelectasis from a high FiO<sub>2</sub>. Breathing 100% O<sub>2</sub> increased the dispersion of blood flow, but neither intrapulmonary shunt nor dispersion of alveolar ventilation increased.<sup>49</sup> These findings were interpreted as release of hypoxic pulmonary vasoconstriction.

In one study,<sup>52</sup> measurements were repeated while the uninvolved (good) right lung was down. Gas-exchange abnormalities in these patients had been previously reported to be more common in the left lung.<sup>52</sup> The improvement of  $\text{PaO}_2$  on moving the patient from supine to the right lateral decubitus position was associated with an improvement in  $\dot{V}_A/\dot{Q}$  relationships but not in shunt, although a beneficial effect on overall gas exchange, secondary to a simultaneous increase of cardiac output, could not be ruled out. These results are at variance with those in patients with unilateral lung disease, in whom either intrapulmonary shunting or  $\dot{V}_A/\dot{Q}$  mismatch improved when the uninvolved (good) lung was down.<sup>47</sup>

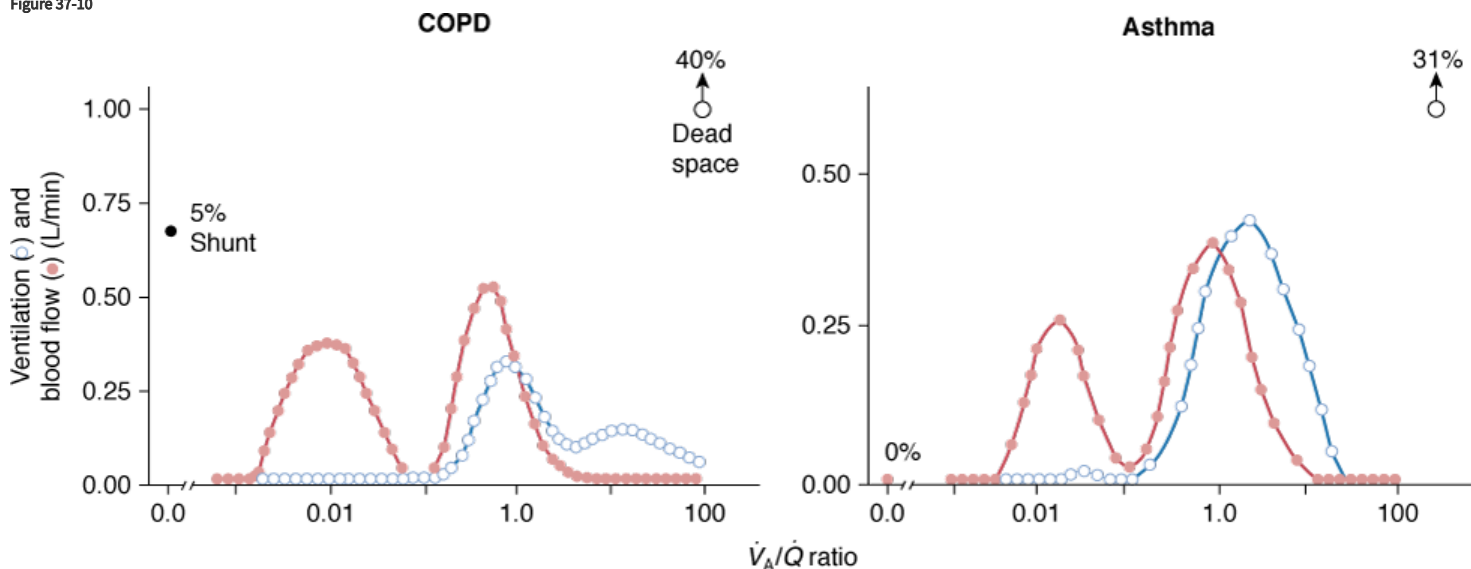
## Chronic Airflow Limitation (“Dry Lung”)

Although exacerbations of COPD and acute severe asthma have a common functional hallmark, chronic airflow obstruction with or without associated reversibility, they show substantial pathophysiologic differences. A contrasting feature of pulmonary gas exchange in patients with “dry lung” and in patients with “wet lung” is that the functional findings observed during exacerbations can be better interpreted in light of the findings observed during the stable state.

## Chronic Obstructive Pulmonary Disease Exacerbations

Patients with COPD show up to four different patterns of  $\dot{V}_A/\dot{Q}$  distributions (Fig. 37-10). One pattern may be characterized by a mode that includes a substantial amount of blood flow diverted to lung units with very low  $\dot{V}_A/\dot{Q}$  regions, known as “low  $\dot{V}_A/\dot{Q}$  mode” (type L).<sup>53,54</sup> Another profile may include alveolar units with high  $\dot{V}_A/\dot{Q}$  regions, named “high  $\dot{V}_A/\dot{Q}$  mode” (type H). This pattern likely suggests that most of the alveolar ventilation is distributed to zones with higher  $\dot{V}_A/\dot{Q}$  ratios. A third profile is a mixed high-low  $\dot{V}_A/\dot{Q}$  mode (type H-L), which consists of additional modes above and below the main body of  $\dot{V}_A/\dot{Q}$  ratios. A fourth pattern has only two broadly unimodal dispersions. All in all, the dispersions of blood flow or ventilation, or both, tend to be severely increased. Intrapulmonary shunting is slightly elevated and dead space is mildly to moderately increased. The minimal amount of shunt suggests that the efficiency of collateral ventilation is very active or that complete airway obstruction does not occur functionally except in a few airways that are completely occluded, possibly by inspissated bronchial secretions.

Figure 37-10



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**Left panel.** Typical bimodal patterns of both blood flow and alveolar ventilation in a representative patient with COPD needing mechanical ventilation. Both shunt and dead space are modestly increased. **(Right panel)** Typical bimodal pattern of blood flow in a representative patient with asthma needing mechanical ventilation. At variance with the COPD patients shunt is absent and dead space is modestly decreased. For further explanation, see text. (Reproduced, with permission, from Rodriguez-Roisin.<sup>16</sup>)

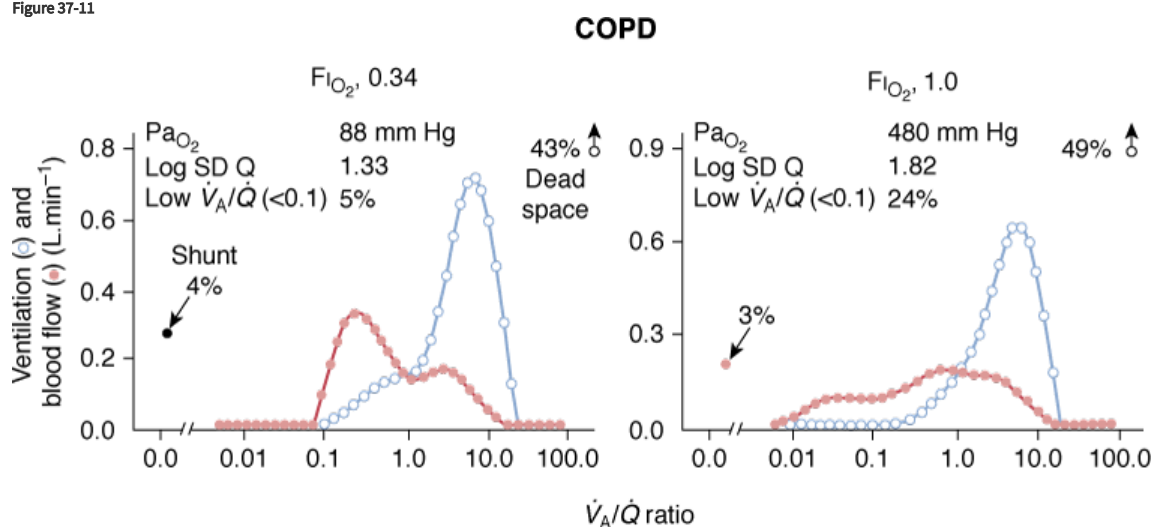
An important concept is that patients with a COPD exacerbation, whether requiring or not requiring ventilator support, exhibit mild amounts of shunt (usually less than 10% of cardiac output) although quantitatively more severe  $\dot{V}_A/\dot{Q}$  patterns than do patients with stable COPD.<sup>53,54</sup> If a patient with an exacerbation of COPD shows moderate-to-severe amounts of shunt (i.e., 20% of cardiac output) despite normal chest radiology—which excludes collapse, consolidation, or edema within the lung—then the possibility of a reopening of the foramen ovale should be considered.<sup>55,56</sup>

**Breathing 100% Oxygen.**

During COPD exacerbations, administration of 100% O<sub>2</sub> rapidly produces full nitrogen washout of alveolar units, even in patients with low or very low  $\dot{V}_A/\dot{Q}$  ratios. A steady state is reached by approximately half an hour, resulting in PaO<sub>2</sub> values close to 500 mm Hg.<sup>21,57</sup> These data indicate that shunt is trivial (less than 10% of cardiac output) or negligible.

Breathing 100% O<sub>2</sub> always worsens  $\dot{V}_A/\dot{Q}$  relationships, as assessed by a significant increase in the dispersion of blood flow (Log SD Q). In addition, there is a discrete decrease of the dispersion of alveolar ventilation, and dead space tends to increase (Fig. 37-11). As discussed above, the further worsening in  $\dot{V}_A/\dot{Q}$  relationships suggests attenuation of hypoxic pulmonary vasoconstriction, even though both pulmonary arterial pressure and vascular resistance remain essentially unaltered. This  $\dot{V}_A/\dot{Q}$  deterioration results in a slight but significant increase in PaCO<sub>2</sub>, which also can be influenced by the Haldane effect. The latter was more strongly demonstrated in patients with COPD exacerbations while spontaneously breathing 100% O<sub>2</sub>.<sup>58</sup> Alternatively, the slight decrease in the ventilation dispersion could reflect some redistribution of blood flow, from areas of high to areas of low  $\dot{V}_A/\dot{Q}$ , induced by the accentuated hypoxic vascular response. The lack of increase of shunt in patients with COPD, unlike in patients with ALI or ARDS, indicates that reabsorption atelectasis does not take place. This suggests that either collateral ventilation is very efficient or regional airway occlusion is never functionally complete.

Figure 37-11



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Effects of breathing 100% oxygen (right panel) in a representative patient with COPD. Compared with low FiO<sub>2</sub> (left panel), PaO<sub>2</sub> increased substantially (≥400 mm Hg) but intrapulmonary shunt remained unaltered; in contrast, the dispersion of blood flow (Log SD Q) and the amount of blood flow distributed to areas with low  $\dot{V}_A/\dot{Q}$  ratios increased markedly. This suggests release of hypoxic vasoconstriction without developing reabsorption atelectasis. Dead space tended to increase. Compare these data with those in patients with ARDS (see Fig. 37-2).

Another striking finding is that when maintenance FiO<sub>2</sub> is resumed, PaO<sub>2</sub> rapidly (in less than half an hour) regains the values observed before the institution of breathing 100% O<sub>2</sub>.<sup>21</sup> This indicates that nitrogen washout of alveolar units rapidly ceases, even in patients with severe airflow limitation, an observation at variance with ARDS (see Fig. 37-5).<sup>21</sup>

**Effects of Positive End-Expiratory Pressure.**

The potential effects of PEEPi (auto-PEEP)<sup>59</sup> on gas exchange should be kept in mind in patients with COPD requiring mechanical ventilation. In these patients, alveolar pressure can remain positive throughout the respiratory cycle despite the absence of PEEP set on the ventilator.<sup>59</sup>

Because low levels of external PEEP may offset the deleterious effects of PEEPi, investigators assessed the effects of both PEEP and PEEPi on  $\dot{V}_A/\dot{Q}$  relationships in patients with chronic airflow obstruction (all but one patient with acute severe asthma had COPD).<sup>60</sup> At low levels of external PEEP (approximately 50% of PEEPi), arterial blood gases improved secondary to an overall optimization of  $\dot{V}_A/\dot{Q}$  relationships without changes in pulmonary mechanics or hemodynamics. With further increases in external PEEP so that PEEP was equal to PEEPi, airway pressures slightly

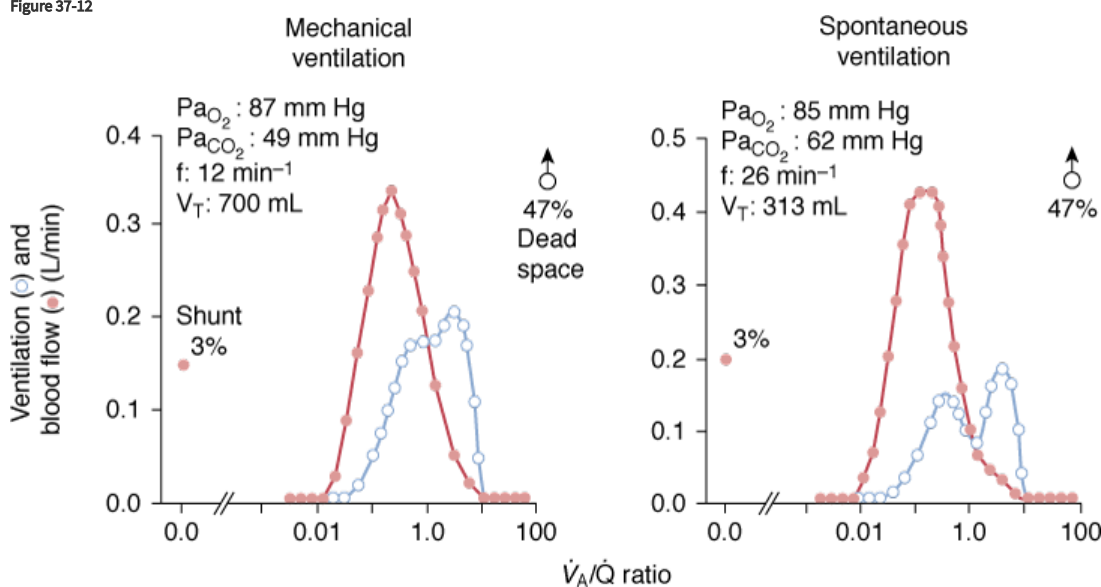


increased without further  $\dot{V}_A/\dot{Q}$  improvement. The investigators also showed that the use of “permissive” or “controlled” hypoventilation<sup>61</sup> to deliberately reduce PEEPi (by 50%) optimized O<sub>2</sub> delivery secondary to a simultaneous increase in cardiac output. Accordingly, the use of low levels of external PEEP together with “permissive” hypoventilation may be advantageous in patients with COPD needing ventilator support. PEEP may improve gas exchange with less risk of barotrauma. These findings have been strengthened by a study<sup>62</sup> in patients with expiratory flow limitation (i.e., COPD and asthma exacerbations) and PEEPi that showed that application of external PEEP below measured PEEPi did not cause hyperinflation or increased intrathoracic pressure, and resulted in no alteration in lung mechanics, hemodynamics, or gas exchange.

#### Weaning.

$\dot{V}_A/\dot{Q}$  relationships in patients with COPD during weaning from mechanical ventilation has been investigated (Fig. 37-12).<sup>57</sup> No major differences in pulmonary and systemic hemodynamics were shown between mechanical and spontaneous breathing, although cardiac output increased significantly when patients were removed from the ventilator. Arterial P<sub>O<sub>2</sub></sub> and O<sub>2</sub> consumption did not change between the two conditions, but mixed venous P<sub>O<sub>2</sub></sub> and O<sub>2</sub> delivery increased significantly when patients were weaned from the ventilator. During spontaneous breathing, minute ventilation remained essentially unchanged, but respiratory frequency increased and tidal volume fell, resulting in rapid shallow breathing, increased P<sub>aCO<sub>2</sub></sub>, and decreased pH.

Figure 37-12



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Effect of ventilator weaning in a representative patient with COPD. Compared with mechanical ventilation (left panel), weaning (right panel) induced rapid and shallow breathing resulting in increased P<sub>aCO<sub>2</sub></sub>. Arterial P<sub>O<sub>2</sub></sub> did not decrease because the simultaneous increase in cardiac output (not shown) when mechanical ventilation was removed tended to improve PaO<sub>2</sub>, the final result being an unaltered PaO<sub>2</sub>. (Reprinted with permission of the American Thoracic Society. Copyright © 2012 American Thoracic Society. Torres A, Reyes A, Roca J, et al. Ventilation-perfusion mismatching in chronic obstructive pulmonary disease during ventilator weaning. *Am Rev Respir Dis.* = 1989;140:1246–1250. Official Journal of the American Thoracic Society. Modified with permission of the American Thoracic Society.)

A considerable increase in the percentage of blood flow to areas of low  $\dot{V}_A/\dot{Q}$  was observed during spontaneous ventilation. In contrast, shunt, dispersion of blood flow, and dead space did not change. Therefore, there was further  $\dot{V}_A/\dot{Q}$  worsening during spontaneous ventilation after removal of ventilator support. This worsening can be explained by alterations in the breathing pattern and also by changes in cardiac output. Yet, PaO<sub>2</sub> did not undergo major changes, indicating that arterial blood gases were not sufficiently sensitive to detect  $\dot{V}_A/\dot{Q}$  changes because other factors, such as minute ventilation and cardiac output were modulating pulmonary gas exchange. Indeed, cardiac output increased substantially after cessation of mechanical ventilation because of a simultaneous abrupt increase in venous return. Similarly, there were increases in mixed venous P<sub>O<sub>2</sub></sub> and O<sub>2</sub> delivery secondary to the increased cardiac output. Nevertheless, the resulting beneficial effect of the increased cardiac output on PaO<sub>2</sub> was counterbalanced by the simultaneous  $\dot{V}_A/\dot{Q}$  worsening, induced in turn by a less-efficient breathing pattern and an increased overall blood flow. Furthermore, using pressure-support ventilation during weaning in patients with exacerbations of COPD prevented worsening

of  $\dot{V}_A/\dot{Q}$  relationships during the transition from positive-pressure ventilation to spontaneous breathing.<sup>63</sup> Gas-exchange and hemodynamic abnormalities were no different between assist-control and pressure-controlled ventilation when both strategies provided similar levels of ventilator assistance.

Similar results, although without further  $\dot{V}_A/\dot{Q}$  worsening, were observed during weaning from mechanical ventilation in patients following coronary artery bypass (half of whom had chronic airflow limitation).<sup>64</sup> Patients with head trauma<sup>48</sup> showed a similar gas-exchange response upon discontinuation of mechanical ventilation, although without associated changes in ventilation and cardiac output.

The key role played by hemodynamic changes has been also emphasized during unsuccessful weaning in patients with COPD.<sup>64</sup> Additional factors contributing to weaning failure include preexisting cardiovascular disease, which may be aggravated by dramatic changes in venous return. An increase of gastric pressure during spontaneous ventilation with subsequent increased splanchnic flow could be an additional mechanism.<sup>65</sup> Oxygen consumption also increases significantly. Inert gas and isotopic studies subsequently revealed that the critical alteration of the ventilation during weaning caused the development of basal regions with very low  $\dot{V}_A/\dot{Q}$  ratios.<sup>66</sup>

The increased metabolic demands in these patients could have further negatively influenced the final  $Pa_{O_2}$ , thus offsetting the positive effects of the increased cardiac output. In patients with COPD, switching mechanical ventilation to either continuous positive-pressure ventilation or pressure support, increased  $O_2$  consumption and decreased minute ventilation, with marked increments  $Pa_{CO_2}$  and no change in  $Pa_{O_2}$ .<sup>67</sup> Conceivably, the simultaneous increase in cardiac output (not measured) prevented a fall in  $Pa_{O_2}$ . Subsequently, it was observed that patients needing mechanical ventilation (mostly with COPD and cardiac disorders) who failed a trial of spontaneous breathing developed a progressive reduction of mixed venous  $O_2$  saturation secondary to the combined effect of a relative decrease in  $O_2$  transport, possibly related to a decreased right-ventricular and left-ventricular afterload, and an increase in  $O_2$  extraction by the tissues.<sup>68</sup>

#### Noninvasive Ventilation.

The short-term effect of noninvasive ventilation for an exacerbation of COPD has been investigated by our group.<sup>69</sup> The beneficial effect on pulmonary gas exchange—decreased  $Pa_{CO_2}$  and increased  $Pa_{O_2}$  and pH—was essentially achieved by a more efficient ventilatory pattern (slower and deeper breathing) without a favorable influence on  $\dot{V}_A/\dot{Q}$  imbalance. A significant decrease in cardiac output during mechanical support, because of increased intrathoracic pressure, did not decrease  $Pa_{O_2}$ .

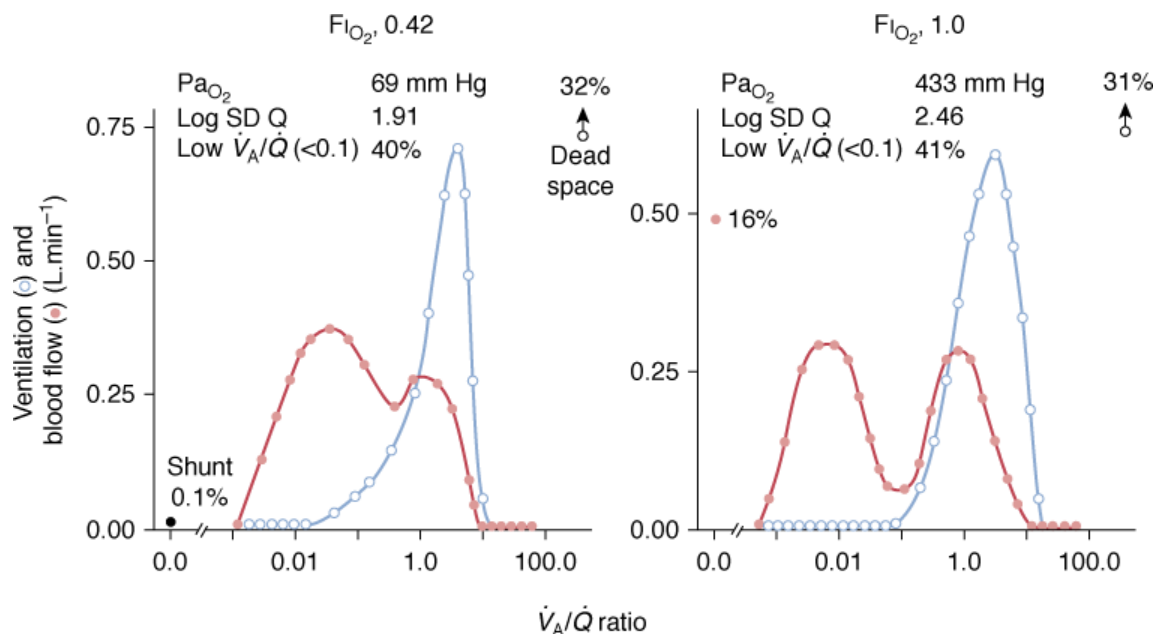
### Acute Severe Asthma

In postmortem studies in patients with sudden fatal asthma, bronchioles reveal increased amounts of luminal occlusion, increased smooth-muscle thickness, and inflammatory infiltrate with both mononuclear cells and eosinophils. Muscular pulmonary arteries adjacent to occluded and inflamed peripheral airways showed a marked inflammatory process in their walls more noticeably when close to airways.<sup>70</sup>

#### Mechanisms of Hypoxemia

Patients with life-threatening status asthmaticus who need mechanical ventilation show the most abnormal  $\dot{V}_A/\dot{Q}$  pattern in patients with asthma: a marked bimodal blood flow profile that may include 50% or more of cardiac output (Fig. 37-13).<sup>71,72</sup> Severe airways obstruction produces regions of low  $\dot{V}_A/\dot{Q}$  that remain perfused but poorly ventilated. This must increase blood flow dispersion more than ventilation dispersion. Conceivably, the high  $FI_{O_2}$ , together with high doses of bronchodilators with potential vasodilating effects, may attenuate hypoxic pulmonary vasoconstriction, hence contributing to the increase in perfusion of areas with low  $\dot{V}_A/\dot{Q}$  ratios. The relatively well-preserved  $Pa_{O_2}$  in the presence of such severe  $\dot{V}_A/\dot{Q}$  mismatch is noteworthy. This can be attributed to the inordinately high levels of both cardiac output and overall ventilation, which reinforce the major role played by the extrapulmonary determinants of  $Pa_{O_2}$  in modulating gas exchange.<sup>5,11,16</sup>

Figure 37-13



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Effect of breathing 100%  $O_2$  (right panel) in a representative patient with acute severe asthma. Compared with low  $F_{I_{O_2}}$  (left panel),  $Pa_{O_2}$  increased substantially ( $\geq 400$  mm Hg) associated with increases in intrapulmonary shunt, the dispersion of blood flow (Log SD Q), and the amount of blood flow distributed to areas of low  $\dot{V}_A/\dot{Q}$  ratios. This suggests the development of reabsorption atelectasis (or selective redistribution of pulmonary blood flow) accompanied with the release of hypoxic vasoconstriction. Compare these data with those seen in patients with COPD (see Fig. 37-12).

Unexpectedly, shunt was negligible or trivial in these patients. Given the presence of abundant, tenacious, and viscous secretions, one would expect alveolar units distal to narrowed airways to be collapsed. This, however, need not be the case. Collateral ventilation and hypoxic pulmonary vasoconstriction are exceptionally efficient. Moreover, airway occlusion is never functionally complete. Another unexpected finding was the presence of normal dead space, the mechanism of which remains unsettled. The coexistence of diffusion limitation for  $O_2$  was also ruled out (Table 37-2).

Table 37-2: Characteristics of Pulmonary Gas Exchange in Chronic Airflow Limitation ("Dry Lung")

|                              | COPD                        | Asthma                     |
|------------------------------|-----------------------------|----------------------------|
| <i>Principal mechanisms</i>  |                             |                            |
| Shunt                        | Mild (<10% of $Q_T$ )       | Absent                     |
| $\dot{V}_A/\dot{Q}$ mismatch | Severe (nonuniform pattern) | Severe (uniform pattern)   |
| $O_2$ Diffusion limitation   | Absent                      | Absent                     |
| <i>100% Oxygen effects</i>   |                             |                            |
| Increase in $Pa_{O_2}$       | Marked ( $\geq 500$ mm Hg)  | Marked ( $\geq 500$ mm Hg) |
| Increase in shunt            | Absent                      | Mild/moderate              |
| Hypoxic vascular response    | Increased                   | Increased                  |

Breathing 100% O<sub>2</sub> induced mild amounts of shunt (less than 10% of cardiac output) in patients with asthma (see Fig. 37-13), suggesting either the presence of critical alveolar units leading to reabsorption atelectasis, or redistribution of pulmonary blood flow of preexisting small shunts, or both.<sup>71</sup> In addition,  $\dot{V}_A/\dot{Q}$  mismatch substantially deteriorated, as shown by a considerable increase of the dispersion of pulmonary blood flow. This suggests that hypoxic pulmonary vasoconstriction was decreased. These data are consistent with the morphologic postmortem findings<sup>70</sup> showing that muscular pulmonary arteries adjacent to occluded peripheral airways had marked inflammatory wall involvement that was not characteristic of that associated with chronic hypoxia. Conceivably, these structural abnormalities may reflect pulmonary vascular leakage related to the release of inflammatory mediators.<sup>73</sup> In contrast, the dispersion of alveolar ventilation decreased significantly. A redistribution of blood flow from areas with units of high to areas with units of low  $\dot{V}_A/\dot{Q}$  ratio is a potential explanation. A tendency toward an increase in dead space, also shown during 100% O<sub>2</sub> breathing, could reflect hyperoxic bronchodilation.

### Effects of Positive End-Expiratory Pressure

The use of PEEP in patients with severe asthma is controversial. Anecdotal reports in patients with life-threatening asthma show conflicting results. Although some studies showed improvement in gas exchange and lung mechanics,<sup>74,75</sup> others found marked increases in PaO<sub>2</sub> associated with gas trapping and increased airway and intrathoracic pressures with decreased cardiac output.<sup>76</sup> From a therapeutic viewpoint, a study by our group<sup>60</sup> of patients with COPD and acute respiratory failure suggests that low levels of PEEP, combined with some degree of “permissive” hypoventilation, might help in optimizing gas exchange in patients with acute severe asthma needing mechanical ventilation.

## Effect of Special Modes of Ventilation

### High-Frequency Ventilation

During the 1980s there was much interest in mechanical ventilation using small tidal volumes combined with very high frequencies. Few techniques of mechanical ventilation have stimulated as much excitement in the minds of investigators as high-frequency ventilation (HFV), resulting in hundreds of research studies. A reduction in complications such as hemodynamic depression and barotrauma associated with conventional modes was invoked as one of the major advantages.<sup>77</sup>

A comparison of HFV and controlled mechanical ventilation (CMV) in a canine model showed that the efficiency of PaO<sub>2</sub> was no different, but the amount of areas of high  $\dot{V}_A/\dot{Q}$  was increased.<sup>78,79</sup> Additional experiments showed that there was an enhanced transport of high-solubility inert gases (approximately twice that for other inert gases) along the airways during HFV. It was suggested that this enhanced transport was dependent on the wet luminal surface of conducting airways.<sup>78</sup>

Although the mechanism of gas delivery by HFV remains to be elucidated, one of the hypotheses is based on augmented dispersion.<sup>80</sup> The theory posits that augmented dispersion during HFV should be less dependent on airway resistance and pulmonary compliance than during CMV. Accordingly, regions of low  $\dot{V}_A/\dot{Q}$  might be better ventilated than during CMV. In a canine model of extensive areas of low  $\dot{V}_A/\dot{Q}$  but little shunt associated with bronchoconstriction and mucus secretion, HFV was no more effective in delivering fresh gas to such areas than was CMV.<sup>81</sup>

### Continuous-Flow Ventilation

Continuous-flow ventilation (CFV) is achieved by delivering continuous streams of gas through cannulas directed down the two main bronchi, in the absence of tidal excursions in lung volume. When CFV was compared to CMV in an animal model,<sup>82</sup> CFV caused significant deterioration in  $\dot{V}_A/\dot{Q}$  matching secondary to an increase in the amount of dispersion of pulmonary blood flow. This suggested that CFV induced a nonuniform ventilation distribution and a redistribution of pulmonary perfusion.

In supine dogs, CO<sub>2</sub> elimination was more efficient with endotracheal insufflation than with tracheal insufflation, but the P(a-a)O<sub>2</sub> was larger during CFV than during CMV irrespective of the type of insufflation.<sup>83</sup> Conversely, elimination of CO<sub>2</sub> and P(a-a)O<sub>2</sub> was lower when prone than when supine. In the prone position, gas distribution was uniform with both modes of ventilation. The increased P(a-a)O<sub>2</sub> when supine during CFV was negatively correlated with the decreased ventilation of the dependent zones of the lung, suggesting further  $\dot{V}_A/\dot{Q}$  inequalities. The improved gas exchange during CFV in dogs lying prone reflects a more efficient  $\dot{V}_A/\dot{Q}$  matching, presumably because the distribution of pulmonary blood flow is nearly uniform.

## Important Unknowns

The lung is a complex system deriving its chief function from the interaction of two very complicated networks that respectively determine regional perfusion and ventilation. It is now well accepted that the normal lung is exquisitely sensitive to gravity, which normally causes regional differences in blood flow, alveolar ventilation, pleural pressures, and parenchymal stress.

The most common model of pulmonary gas exchange proposes a gravity-induced vertical distribution of perfusion and ventilation, and regional matching of ventilation and perfusion is accomplished primarily by the shared effects of gravity. Radioactive gases and chest wall scintillation counters were used by a number of investigators in the early 1960s to estimate regional ventilation and perfusion within anterior–posterior cores of lung tissue.

Despite being quantitative, studies using MIGET could not identify the topographical distribution or size of low  $\dot{V}_A/\dot{Q}$  units or determine the extent to which the distribution of pulmonary perfusion changes in different experimental and clinical interventions. There have been, however, a few complex attempts to interpret pulmonary perfusion heterogeneity within the context of the gravitational model as random noise. Fractal analysis, a revolutionary mathematical science, demonstrated that this “error” can be analyzed as a fundamental property of the biologic system. Fractal analysis permits characterization of processes or structures that are not easily represented by traditional analytical tools and provides insights and tools for constructing new models. The primary incentive for developing a fractal model of pulmonary blood flow distribution was to explore a unifying mechanism that explains the heterogeneity of perfusion as well as the observation that neighboring lung regions have similar magnitudes of flow.<sup>84</sup>

Another approach has been the use of positron emission tomography imaging of intrapulmonary kinetics of intravenously infused tracer nitrogen-13 ( $^{13}\text{NN}$ ) to measure the regional distribution of ventilation and perfusion.<sup>85</sup>

Ideally, one would like to have a tool to timely measure both ventilation and perfusion taking into account many of the alluded-to caveats. This should include the overcoming of the great difficulties of achieving real steady-state conditions while invasiveness is minimized to [cope](#) with cost and time investments.

## The Future

Until the development of some completely new noninvasive, simple technique, it is difficult to anticipate what the future holds for research on pulmonary gas exchange. Over the past three decades, we have witnessed hundreds of elegant studies, of both an experimental and clinical nature. These studies have extensively, and complementarily, unravelled the basic mechanisms of gas exchange across a spectrum of acute and chronic respiratory conditions. Key to the gains in knowledge has been parallel advances that have broadened and deepened our understanding of the clinical management of these conditions. Given this high level of research excellence, it is difficult to predict what will come next. Nevertheless, several worthwhile questions remain to be addressed, particularly in the area of clinical management.

One recent MIGET study was undertaken in patients experiencing a severe exacerbation of COPD,<sup>86</sup> and it addressed the gas-exchange response to a common therapy. It has long been recognized that nebulized, short-acting  $\beta_2$ -agonists can induce mild arterial hypoxemia in patients with COPD, but most of the published work was conducted in the setting of stable COPD. The new study, conducted in patients experiencing a severe exacerbation of COPD requiring hospitalization, revealed no significant gas-exchange abnormalities after use of nebulized salbutamol. During convalescence, however, the same patients experienced small decrements in  $\text{PaO}_2$ , secondary to  $\dot{V}_A/\dot{Q}$  worsening, on administration of a  $\beta_2$ -agonist. This study addressed not only the salbutamol-induced interplay between extrapulmonary (ventilatory, metabolic, and vascular effects) and intrapulmonary mechanisms (consequences of  $\dot{V}_A/\dot{Q}$  mismatching), providing a better insight into the role of gas-exchange pathophysiology in the management of COPD exacerbations. It further confirmed that short-acting bronchodilators are safe when used in the therapy of life-threatening exacerbations of COPD.

## Conclusion

Arterial  $\text{P}_{\text{O}_2}$  and  $\text{P}_{\text{CO}_2}$  are the end-point outcomes of the gas-exchange state and are governed by the interplay of several intrapulmonary and extrapulmonary components. The most remarkable intrapulmonary factors are  $\dot{V}_A/\dot{Q}$  mismatching and intrapulmonary shunt; in contrast, diffusion limitation to  $\text{O}_2$  plays a marginal role. Among the extrapulmonary factors, inspired  $\text{P}_{\text{O}_2}$ , overall ventilation, cardiac output, and  $\text{O}_2$  consumption are viewed as the most influential.

While breathing 100% O<sub>2</sub>, despite substantial PaO<sub>2</sub> improvements,  $\dot{V}_A/\dot{Q}$  relationships worsen characteristically according to the underlying nature of acute respiratory insufficiency. In patients with “wet lung” (ALI or ARDS), shunt increases secondary to the development of reabsorption atelectasis without release of hypoxic pulmonary vasoconstriction. In patients with “dry lung” (COPD or asthma), hypoxic vasoconstriction is ultimately released while shunt remains unchanged. Gas-exchange abnormalities may be influenced by changes in cardiac output depending on different ventilator modalities, in particular during weaning from mechanical ventilation.

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